

Vitreaclear: Reducing Vision Disturbance From Floaters

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Abstract

Eye floaters are small shadows or specks that drift across the visual field, caused by clumps of collagen fibers within the vitreous humor, the gel-like substance filling the eye. These disturbances are common, affecting up to 60% of individuals over 60 and those with conditions such as myopia or Posterior Vitreous Detachment (PVD). For many individuals, floaters are more than a minor inconvenience; they interfere with essential daily activities such as reading, driving, and recognizing faces, significantly diminishing quality of life. Patients often express frustration, anxiety, and a strong desire for relief, with some willing to accept substantial risks for treatment. This widespread prevalence and impact create a pressing need for effective solutions. The current standard of care offers limited options, many of which fail to meet the needs of patients. Observation, the most common approach, provides no symptom relief and leaves patients to endure ongoing visual disturbances. Pars plana vitrectomy (PPV), an invasive surgical procedure, effectively removes floaters but comes with serious risks, including retinal detachment, cataracts, and infections. These complications, along with the extended recovery period, make PPV a last resort for severe cases, excluding many patients who might otherwise benefit from treatment. Laser vitreolysis, a less invasive option, has shown some promise but remains inconsistent in its outcomes and carries risks of collateral damage to ocular structures. Pharmacological approaches are still in early stages of development, with no proven or widely available therapies. **A way to safely and precisely reposition visually disruptive vitreous floaters for myodesopsia patients so that they can achieve improved vision without undergoing invasive surgery or risking retinal complications** is urgently needed. Our team has developed VitreaClear, an ultrasound-based acoustic repositioning system using precisely arranged piezoelectric transducers to generate controlled acoustic wave fields that create pressure nodes capable of displacing microparticles in a low-impedance media. Current work focuses on transitioning from single-element to phased array configurations. Next steps include expanding diagnostic tools for comprehensive force, power, temperature, and safety parameter monitoring, conducting tests in viscous fluids simulating vitreous environment, and progressing toward cadaver eye validation and clinical trials.

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1. Significance

1.1 Unmet Health Need

Eye floaters are a prevalent ocular condition that affects approximately 60% of individuals over the age of 60 years [2]. The incidence appears to be higher in women compared to men and is notably more common among patients with myopia, with around 60% of eye floater patients in 2011 study by Khoo Teck Puat Hospital in Singapore also experiencing Myopia, with a further third of those patients rating as high in Myopia [4]. Additionally, controls in the study (300+ patients, 21 and above) found that the utility value lost due to eye floaters does not change with the presence or absence of myopia and other conditions, leading credence to the idea that eye floaters directly cause the measured quality of life loss.

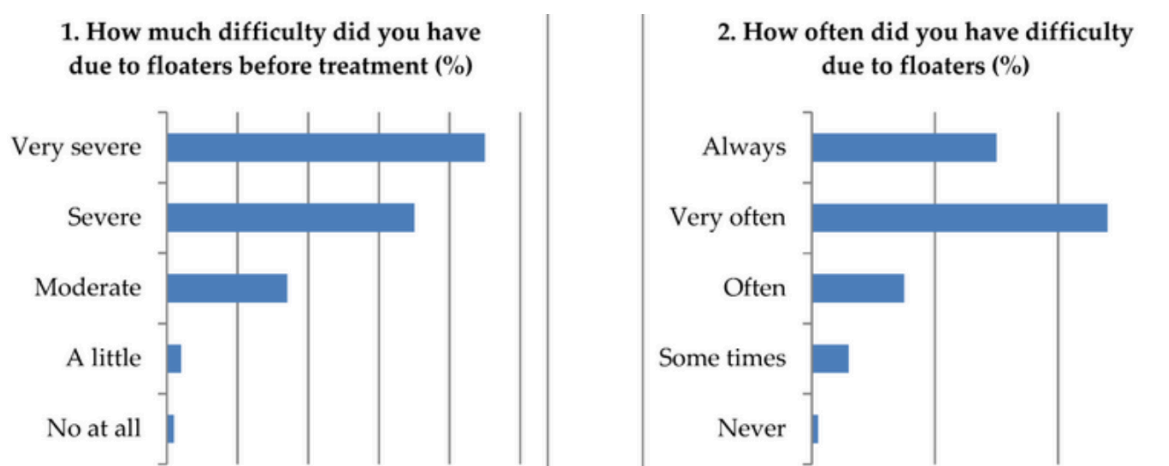


Figure 1. Patient-Reported Challenges with Eye Floaters (n=224): **Graph 1** (left) shows the distribution of perceived difficulty caused by floaters, while **Graph 2** (right) illustrates the frequency of difficulty experienced by patients [4].`

The phenomenon occurs when collagen fibers within the vitreous sac detach due to either natural gel liquefaction, Posterior Vitreous Detachment (PVD) or trauma, resulting in shadows cast on the retina. Patients commonly report observing gray shapes, web-like patterns, or, in more severe cases, a Weiss ring, which manifests as a dark ring in the visual field and is associated with the deterioration of collagen fibers near the optic nerve after PVD [3], which is the main cause of eye floaters.

Epidemiological and Economic Impact

Although many individuals acclimate to floaters without seeking treatment, these visual disturbances can significantly impair daily activities, including reading and driving

[3]. According to a survey of 311 patients at Khoo Teck Puat Hospital in Singapore, eye floater patients would be willing to accept substantial risks—such as an 11% chance of death or a 7% chance of permanent blindness—in exchange for relief from their symptoms. These risk values are similar to values obtained in utility studies on other eye conditions such as macular degeneration, diabetic retinopathy, and myopia in cases of similar severity [4]. Floaters are often perceived as less serious than other visual conditions, however, analysis on quality of life shows that eye floaters have serious shortfalls in treatment options given their utilitarian impact.

Furthermore, using data from the Bureau of Labor Statistics and the CDC, the economic burden of visual impairment in the United States is estimated at approximately \$16.33 billion annually due to reduced labor force participation [5]. While this figure encompasses a range of conditions, including nearsightedness, farsightedness, and cataracts—each of which has well-established treatment options—therapies specifically targeting floaters are still in their infancy, with initial safety testing for procedures like laser vitreolysis having commenced only in 2016 [4]. This underscores the urgent unmet need for continued research and development of effective, noninvasive treatments to enhance both quality of life and economic productivity for those affected by this condition.

Stakeholder Analysis

For this issue, the targeted stakeholders consist of ophthalmologists, optometrists, and patients. Ophthalmologists and optometrists are at the forefront of diagnosing and managing the condition, facing the challenge of balancing treatment risks with patient expectations. Patients with symptomatic floaters often experience significant psychological distress due to persistent visual disturbances, which can affect quality of life and daily functioning. Studies highlight their willingness to accept substantial treatment risks to alleviate symptoms, underscoring the severity of their condition and the inadequacy of existing solutions. Stakeholders are looking for treatments that are safe, effective, minimally invasive, and capable of delivering rapid relief.

For healthcare providers, there is a pressing need for improved diagnostic tools to quantify the impact of floaters on contrast sensitivity and visual function. For patients, the greatest "gain" would be a non-invasive, low-risk intervention that effectively resolves symptoms without the long-term complications associated with current surgical options.

1.2. Existing Solutions

Eye floaters are commonly regarded as benign in clinical settings, and are often treated by observation alone. While observation and monitoring are important to tracking

vitreous floaters on a patient-by-patient basis, recent studies have demonstrated that vitreous floaters cause significant decreases to quality of life [8].

The current standard of care for symptomatic vitreous floaters includes three primary approaches: vitrectomy, YAG laser vitreolysis, and patient adaptation. Vitrectomy is considered the most effective but is also the most invasive, involving surgical removal of the vitreous gel to eliminate floaters. Despite its efficacy, it carries risks such as retinal detachment, cataract formation, and other complications, making it a last-resort option for patients with significant visual impairment [7]. YAG laser vitreolysis, while less invasive, has shown mixed results, with studies reporting limited improvement in symptoms and occasional adverse events such as retinal tears or glaucoma [7].

The third and most common approach is advising patients to adapt to their condition. For the majority of individuals, while floaters are highly annoying and can significantly impact their quality of life, they are not life-threatening. Patients are often encouraged to allow their brain to adjust and "tune out" the floaters over time. This strategy is particularly prevalent because many patients assume they have no other options and choose to ignore their symptoms. However, there are no clear statistics on the proportion of patients following each approach, as many cases go unreported or untreated, further complicating efforts to assess the efficacy and accessibility of treatment options.

In situations where treatment is deemed necessary, several options are available at different levels of effectiveness and invasiveness. The most invasive and effective treatment for vitreous floaters is transconjunctival sutureless pars-plana vitrectomy (PPV), which is conducted by inserting a small instrument into the eye, removing the vitreous down to the base, including any detached vitreous, and finally inserting a replacement gel into the eye. A major risk for vitrectomy operations are retinal breaks, which can be treated post-operation with cryotherapy or laser treatment [7]. There are several additional post operational complications that can occur such as endophthalmitis or cataracts, which can be mitigated with additional precautions, intensive postoperative monitoring, and follow up operations [8].

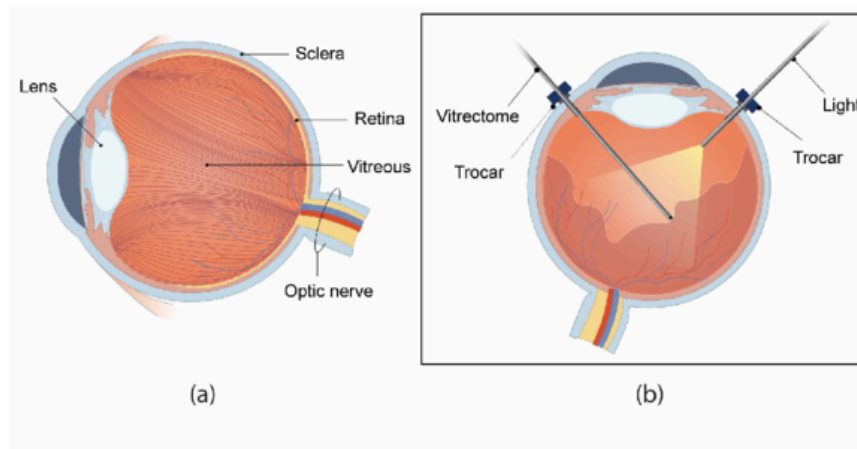


Figure 2. Vitrectomy surgery. (a) Schematic cross-section of the anatomy of the eye, showing the orientation of the collagen fibre network in the eye. (b) Schematic of the set-up during a typical vitrectomy [14].

Other treatments available include laser vitreolysis, which uses a Neodymium YAG laser to specifically target and break apart small clumps of vitreous. This procedure renders the floaters small enough to become imperceptible [6]. Laser treatment is generally considered to be significantly less effective compared to vitrectomy, since some studies have shown figures as low as 35.8% of patients seeing moderate improvements, and only 2.5% seeing significant improvements[7]. However, some of the early research into YAG Lasers from 1993 show significant benefits to the use of laser vitreolysis [6].

While pharmacological interventions to create posterior vitreous detachments have been investigated, there are no current drugs with any proven efficacy for the treatment of vitreous floaters [9]. Research into pharmacological vitreolysis has shown promise, but lack of understanding of molecular mechanisms behind vitreolysis has led to problems in drug development for the destruction of floaters. Currently, very few clinical trials investigating the topic have been performed and further research is necessary, however, there is hope that future developments will yield beneficial contributions to the treatment of eye floaters [10].

Table 1. Current solutions in the market

Solution	Description	Benefits	Drawbacks
Pars-plana vitrectomy	Removal and replacement of vitreous via hollow needle.	Highly effective	Postoperative complications, invasive

Laser vitreolysis	Use of laser to break up vitreous macromolecules.	Somewhat effective, low invasiveness	Limited reliability and effectiveness
Pharmacological vitreolysis	Several different methods usually involve breakdown of vitreous macromolecules by chemical/biological interactions.	Minimally invasive and potentially effective	Currently, there are no available pharmaceutical solutions to vitreous floaters

1.3. Need Statement

The clinical need centers around the development of a non-invasive solution to safely and precisely reposition visually disruptive vitreous floaters in myodesopsia patients, ultimately improving their vision without the need for invasive surgery or exposing them to retinal complications. This latest iteration reflects insights gained from ongoing research, expert feedback, and valuable guidance from our mentor, Scott Lipsky. With a deeper understanding of the risks, challenges, and patient demographics involved, we have refined our need statement to more appropriately align with both the current clinical landscape and the needs of this patient population.

Our focus was then moved on to developing a method to safely ablate symptomatic vitreous floaters in vitrectomy candidates. The goal was to offer a solution that reduced the necessity for invasive surgery and mitigated associated risks, such as retinal detachment and cataracts. At the start of this quarter, following input from our mentor, Scott Lipsky, our team unanimously agreed to shift focus from ablation to repositioning floaters. This change aims to minimize retinal complications while providing a broader, more adaptable solution. Repositioning techniques present a safer alternative with the potential to address a variety of floater profiles without the risks inherent to ablation, such as damage to retinal structures.

Our needs-finding and stakeholder analysis (refer to Appendix A) revealed significant risks associated with invasive treatments, especially for patients who have already undergone vitrectomy. These risks include complications like retinal detachment, cataracts, and glaucoma. To incorporate the broader needs of myodesopsia patients—those experiencing visual disturbances due to vitreous floaters—and focuses on delivering a safe, non-invasive solution that minimizes the risk of retinal complications, our need statement evolved into the following:

"A way to safely and precisely reposition visually disruptive vitreous floaters for myodesopsia patients so that they can achieve improved vision without undergoing invasive surgery or risking retinal complications."

The need statement can be broken down into three parts:

- **The Problem:** Myodesopsia patients suffer from visually disruptive floaters that significantly impair their quality of life. Current treatments, including invasive surgeries like vitrectomy, carry substantial risks, such as retinal detachment, cataracts, and glaucoma.
- **The Population:** The primary target population consists of myodesopsia patients, including those who are experiencing visually disruptive floaters. This group includes:
 - Vitrectomy candidates, who are considering surgery due to the severity of their symptoms.
 - Patients ineligible for vitrectomy, either due to medical conditions or other contraindications.
 - Patients unwilling to undergo existing invasive treatments, who reject the risks associated with surgical interventions.
- **The Outcome:** Provide a non-invasive, safe, and effective solution that can reposition floaters with precision, offering a practical alternative to surgery, thus improving patient vision while reducing the risks associated with current treatments.

This iteration of the need statement is designed to address the critical requirements of the most severe cases—specifically, vitrectomy candidates—while also demonstrating the potential to benefit a broader spectrum of individuals experiencing less severe symptoms. By expanding the scope, this approach extends the treatment's applicability across a larger population of myodesopsia patients.

Our development objectives center on creating a device that has usability across diverse patient profiles to maximize its clinical impact and accessibility. Through iterative development and continuous stakeholder collaboration, our team is committed to delivering an innovative, non-invasive solution aimed at addressing the needs of patients managing floaters that affect their quality of life to varying extents.

1.4. Critical to Customer

The following CTCs represent the requirements necessary for satisfying our stakeholders' needs and expectations. These requirements are defined by our customers—patients, ophthalmologists, and optometrists.

High Priority CTCs (Core Function Requirements)

Reduction of Floater Disturbance - This function addresses visually disruptive floaters by repositioning them to restore visual clarity. This core function aims to deliver immediate benefits to the patient by reducing visual disruptions. Regaining visual clarity can significantly improve the patient's overall well-being and quality of life. The device must effectively reduce floater disturbances to improve myodesopsia patients' quality of life, aligning with the clinical need for a precise and reliable solution. In terms of stakeholder values, patients value improved vision and quality of life; healthcare providers value treatment efficacy and patient satisfaction.

Non-Invasiveness - This core function directly addresses the need for safer, non-invasive alternatives to surgical solutions (i.e. vitrectomy). The concept/solution would effectively target visually disruptive floaters through non-invasive techniques to reduce surgical risks and alleviate patient anxiety. This function is supported by stakeholder feedback prioritizing non-invasive methods to enhance patient safety. Current treatment options are limited to two extremes: no intervention or highly invasive procedures. In terms of stakeholder values, patients value reduced surgical risk and anxiety; Healthcare providers value expanded treatment options and reduced liability.

No/Minimal Side Effects - The treatment must cause no or minimal side effects to ensure patient safety while minimizing potential retinal complications. This function ensures a safe, effective, non-invasive solution with minimal risks, promoting adoption by reducing side effects. The solution offers a reliable, low-risk treatment with minimal side effects, boosting patient confidence and encouraging wider acceptance. Research, along with feedback from both patients and clinicians, emphasizes the importance of minimizing side effects, especially in a sensitive area like the eye. In terms of stakeholder values, patients value safety and peace of mind; healthcare providers value reduced malpractice risk and patient trust.

Expected Requirements

Patient Comfort - Emphasize both physical and psychological comfort to minimize patient stress throughout treatment. This function prioritizes patient comfort and acceptance, promoting adherence to non-invasive treatments and reducing the need for invasive procedures. The concept/solution would increase patient satisfaction and reduce anxiety, leading to a positive overall treatment experience and improved compliance with non-invasive methods. Stakeholder feedback underscores patient comfort as key to accepting non-invasive solutions, with interviews revealing sensitivity to invasive eye

procedures and concerns about discomfort. In terms of stakeholder values, patients value comfortable treatment experience; healthcare providers value patient compliance and satisfaction.

Real-Time Feedback - Integrate monitoring and feedback mechanisms for real-time evaluation of treatment efficacy. This function provides a safety mechanism for quick response to complications, ensuring a reliable and trustworthy treatment process for both patients and clinicians. The solution enhances safety and reliability, boosting patient confidence and providing clinicians with precise control and monitoring. Interviews with optometrists emphasized monitoring as a primary action in the diagnosis and treatment process. Furthermore, clinical priorities for risk mitigation stress the importance of monitoring. In terms of stakeholder values, healthcare providers value precision control and safety monitoring; patients value transparency and confidence in treatment.

Desired Requirement

Efficient Post-Procedural Monitoring - Implement a structured follow-up protocol for ongoing monitoring. This function ensures long-term success by carefully monitoring high-risk patients with symptomatic eye floaters after treatment. We are finalizing a design that integrates smoothly into clinical workflows, offering ease of use, quick operation, and accurate detection to help clinicians monitor changes and support long-term outcomes. Clinical guidelines emphasize the need for continuous monitoring in the treatment of eye floaters. Therefore, our solution should prioritize monitoring to enhance treatment outcomes. In terms of stakeholder values, healthcare providers value workflow integration and long-term patient outcomes; healthcare systems value operational efficiency.

The baseline needs represent core function requirements that must be achieved for any viable solution. These include reduction of floater disturbance, non-invasiveness, and minimal to no side effects, as these directly address the clinical need and safety concerns identified by all stakeholders. Performance needs include functions that enhance the treatment experience and clinical utility, specifically patient comfort and real-time feedback capabilities. Desired requirement represents advanced features that could differentiate our solution in the marketplace, primarily efficient post-procedural monitoring that integrates seamlessly into clinical workflows.

The core functions effectively address the refined need statement. Specifically, a non-invasive and effective solution supports the goal of avoiding invasive surgical intervention by providing an alternative treatment method. Comprehensive real-time monitoring and post-treatment protocols allow for continuous safety verification throughout the treatment process. Finally, by focusing on patient comfort, we aim to

optimize treatment outcomes while maintaining a focus on patient well-being. We are actively refining and developing concepts so that the synergy of these core functions can effectively address our need statement.

1.5 Summary of Significance

Eye floaters affect approximately 60% of individuals over age 60, with patients willing to accept substantial risks—including an 11% chance of death or 7% chance of permanent blindness—for symptom relief, demonstrating the severity of their impact on quality of life. The economic burden of visual impairment reaches approximately \$16.33 billion annually in the United States, yet therapies specifically targeting floaters remain in their infancy.

Existing procedures, devices, and methods do not adequately address this challenge for the following reasons: current treatments are limited to highly invasive vitrectomy with significant complications, moderately effective laser vitreolysis with limited reliability, or patient adaptation with no active intervention. This leaves a critical gap between no treatment and high-risk surgical intervention.

The critical factors that must be addressed to tackle this challenge include: developing non-invasive treatment alternatives, ensuring minimal side effects and complications, providing effective reduction of visual disturbances, maintaining patient comfort throughout treatment, implementing real-time monitoring capabilities, and establishing efficient post-procedural follow-up protocols. These factors collectively support the development of a safe, precise, and non-invasive solution to reposition visually disruptive vitreous floaters for myodesopsia patients.

2. Innovation

2.1 Technical Design Specifications

To translate our Critical to Customer (CTC) requirements into quantifiable engineering specifications, we have developed the following design specifications.

- **Non-Invasive:** The solution uses focused ultrasound energy to reposition floaters. This eliminates the risks of vitrectomy and the collateral damage concerns of YAG laser treatments. Ultrasound is a known non-invasive technology used for many medical problems, thus fitting this specific CTC.
- **Minimal Side Effects:** Unlike vitrectomy, which carries substantial risks (e.g., retinal detachment, cataracts), or YAG laser, which risks retinal damage, ultrasound offers a safer, targeted alternative. Additionally, the solution will follow the requirement of

50 W/cm² as the max power output as outlined by the FDA for ultrasound technologies.

- **Live Feedback During Operation:** Integrate real-time feedback systems would be critical for accurate floater repositioning. This would offer control unavailable in passive adaptation or YAG laser.
- **Minimal Set-Up and Prep Work:** Unlike YAG laser (which requires precise calibration) or vitrectomy (which requires surgical preparation), we are designing our ultrasound treatment to require minimal preparation.
- **Efficient Patient Follow-Up:** With no incisions and minimal risk, post-treatment monitoring is simplified compared to the intensive postoperative care required for vitrectomy or the repeated visits for YAG laser refinement.

The non-invasive, precise, and efficient nature of acoustic repositioning improves patient outcomes compared to the lengthy recovery of vitrectomy and the limited effectiveness of passive adaptation. Unlike vitrectomy (high discomfort) and YAG laser (moderate discomfort), ultrasound treatment is completely non-invasive and requires no incisions or laser ablation. Therefore, using an ultrasound-based system offers a truly non-invasive, precise, and patient-friendly alternative to existing treatments. It has the potential to significantly reduce treatment discomfort, recovery time, and risks associated with surgical interventions, therefore addressing the limitations of current floater treatments.

2.2 Solution

In alignment with the core functions and need statement, the team has identified three potential design solutions to address the challenge of managing eye floaters:

1. **Automated YAG Laser Vitreolysis:** Leveraging computer vision technology to automatically detect and target floaters, seamlessly integrating this functionality with existing laser treatments.
2. **Virtual Reality (VR) Therapy:** Developing VR-based therapeutic games aimed at accelerating the brain's adaptation to floaters.
3. **Acoustic Repositioning of Floaters:** Using precisely arranged piezoelectric disks to generate controlled acoustic wave fields, creating pressure nodes that capture and reposition floaters to non-disruptive areas of the eye.

To evaluate these concepts, the team employed a Quality Function Deployment (QFD) matrix. This approach assessed each solution against existing treatments by analyzing their performance based on Critical-to-Customer (CTC) and Critical-to-Quality (CTQ) metrics. A

simplified version of the QFD matrix is presented in the table on the next page, with a more comprehensive analysis included in Appendix C.

Table 3. Simplified QFD Matrix with darker shades associated to better performance for correlated metrics

	Monitoring and Passive Adaptation	YAG Laser Treatment	Vitrectomy	Automated YAG Laser Treatment	Floater Dissolving Eye Drops	VR Therapy	Acoustic Repositioning
Patient Comfort							
Non-invasive							
No/Minimal Side Effects							
Live Feedback During Operation							
Patient Satisfaction							
Minimal Set-up and Prep Work							
Efficient Patient Follow-up							

Based on comprehensive evaluation through QFD analysis, stakeholder feedback, and technical feasibility assessment, we selected ultrasound-based acoustic repositioning as our leading design solution. This approach uses precisely arranged piezoelectric transducers to generate controlled acoustic wave fields that create pressure nodes capable of capturing and repositioning vitreous floaters to non-disruptive areas of the eye.

Leading Design Description and Function

Vitreaclear's leading design is a wearable ultrasound-based system that uses acoustic radiation force to reposition vitreous floaters away from the central visual axis. This method leverages a multi-element phased array of piezoelectric transducers, currently operating at a frequency of 1 MHz, arranged around the eye in a semi-circular or conformable housing.

The device generates standing acoustic wave fields in the vitreous body. These wave fields form pressure nodes and antinodes, which are capable of exerting controlled forces on floaters suspended in the vitreous medium. By modulating the phase and amplitude across the transducers, the system can steer these pressure zones in 2D (with future development toward 3D), enabling precise, non-contact floater repositioning.

This design was selected as the leading solution after comprehensive evaluation using a QFD matrix, in which it outperformed alternatives such as VR-based neural adaptation therapy and automated YAG laser ablation. Key selection factors included:

- **High effectiveness** in particle manipulation (validated in air and water testing)
- **Non-invasive nature** (external application, no surgical or laser intervention)
- **Low side-effect profile**, adhering to FDA ultrasound safety standards
- **Scalability** for integration into clinical workflows and home-use potential

Vitreaclear directly addresses the identified Critical to Customer (CTC) requirements:

- **Non-invasiveness:** No penetration into the eye, reducing anxiety and risk
- **Floater disturbance reduction:** Repositioning floaters to visually irrelevant regions
- **Minimal side effects:** Maintains safe thermal and mechanical indices under FDA guidelines
- **Patient comfort:** Painless, wearable design with minimal setup
- **Monitoring potential:** Future versions will integrate imaging or tracking feedback

Early Prototype Development

Prototype development has followed an iterative, medium-escalating approach, beginning with air-based setups and progressing toward water-based systems that more accurately simulate the eye's vitreous environment. Initial testing involved single-element piezoelectric transducers operating at approximately 1 MHz. These devices were validated using a vector network analyzer (AIM 4170D) at the Center for Industrial and Medical Ultrasound. Force outputs ranged from 0.010 g to 0.043 g at power levels between 1.43 W and 4.04 W, with consistent floater manipulation observed at 3.7 W. Acrylic tanks were used with deionized and degassed water, polystyrene beads, and gel pads to simulate floaters and enhance impedance matching. Stable power delivery and measurement accuracy were supported using BNC terminal wiring and coaxial cables, which proved superior to taped connectors in minimizing reflection and improving signal clarity.

Further refinements were made after another set of tests. The focal length of the transducer was confirmed to be approximately 2 cm. Calibration of the device was performed using Open-Short-Load (OSL) methodology to correct for systematic measurement errors and ensure accurate impedance characterization. This calibration involved connecting the cable end to three known terminations—open circuit, short circuit,

and 50 Ohm resistor—to establish a reliable reference plane. Calibration was critical to minimize reflected power and ensure optimal transducer performance.

Key testing observations included:

- Medium damping was observed when the transducer was submerged in water, consistent with increased acoustic impedance and viscosity relative to air.
- Epoxy-only matching layers yielded the lowest reflected energy, outperforming both cyanoacrylate and epoxy + alumina configurations.
- Standing wave ratio (SWR) approached optimal impedance matching at 1.001 and 1.007 MHz for the two transducers respectively, indicating minimal reflected power and better acoustic coupling.
- Device wiring was critical: screw-in terminals with electrical tape introduced more variation, while direct BNC coaxial connections improved signal stability. Future setups should prioritize soldered BNC connections for optimal reliability.

Design Enhancements Underway:

- Exploring PDMS encapsulation for transducer waterproofing and long-term durability in liquid media.
- Prototyping a phased-array and control board to enable directional beam steering and phase delay for acoustic tweezers.
- Evaluating glycerin-water, agarose, and collagen solutions to better mimic the damping and viscosity of the vitreous humor.

Next steps:

- Phase 2: Demonstration of particle manipulation in viscous vitreous model.
- Phase 3: Closed-loop 2D/3D phased-array control for dynamic steering
- Phase 4: Testing on enucleated porcine eyes for anatomical relevance

Stakeholder Analysis

Stakeholder evaluation of our design concept reveals strengths in addressing current treatment limitations. An effective, non-invasive approach expands current treatment options without introducing surgical risks. In the meantime, identified weaknesses include the technical complexity of the acoustic system and the need for extensive clinical validation. The novel nature of acoustic floater manipulation requires comprehensive safety and efficacy studies before clinical implementation. In addition, regulatory approval through the De Novo pathway presents additional timeline and resource challenges.

3. Business and Market Opportunity

3.1 Value Proposition

Medical practitioners, particularly optometrists and ophthalmologists, face significant challenges in treating eye floaters. The Value Proposition Canvas provides a comprehensive framework for understanding the critical needs and expectations of these healthcare professionals when considering new treatment options for their patients.

The jobs to be done for medical practitioners fundamentally center on delivering patient care while maintaining professional credibility and expanding treatment capabilities. Functionally, ophthalmologists and eye care clinics require a treatment method that is validated and demonstrably effective in addressing symptomatic eye floaters.

The pains that medical practitioners seek to eliminate are multifaceted for the following reasons. Current treatment options for eye floaters are limited, invasive, and risky, as surgical procedures inherently involve risks. These limitations constrain practitioners' ability to provide care. There is a lack of safe, non-invasive, or minimally-invasive alternatives for treating eye floaters. Moreover, practitioners bear the responsibility of recommending procedures that may involve significant medical risks. From our team's interview with an optometrist, we learned that at the current stage, monitoring is the primary approach for treating eye floaters.

In terms of gains, practitioners are seeking a safe and effective method to reduce visual disturbance caused by floaters. They desire a solution that provides tangible advantages across several dimensions, namely patient safety and treatment efficacy. A breakthrough treatment would allow them to offer patients a proven alternative to existing invasive procedures, thereby reducing procedural risks, enhancing patient experience, and ultimately improving their patients' quality of life.

The proposed solution effectively meets these professional needs for several reasons. Unlike existing treatments, which rely on invasive surgery (vitrectomy) or laser energy (YAG laser), our proposed solution offers a non-invasive, ultrasound-based approach that safely and precisely repositions floaters. Consequently, our technology reduces complications such as retinal damage, cataract formation, and post-operative recovery time. With a solution that applies acoustics in a novel way, our approach mitigates the significant drawbacks of current treatment options, such as vitrectomy, which is highly invasive. In addition, our technology has the potential to empower medical practitioners by providing a way to effectively manage floaters without invasive intervention, thereby

enabling access to a broader patient population. Consequently, this product can help practitioners overcome existing clinical limitations, expand their treatment options, and deliver more effective patient care.

Value Proposition Canvas

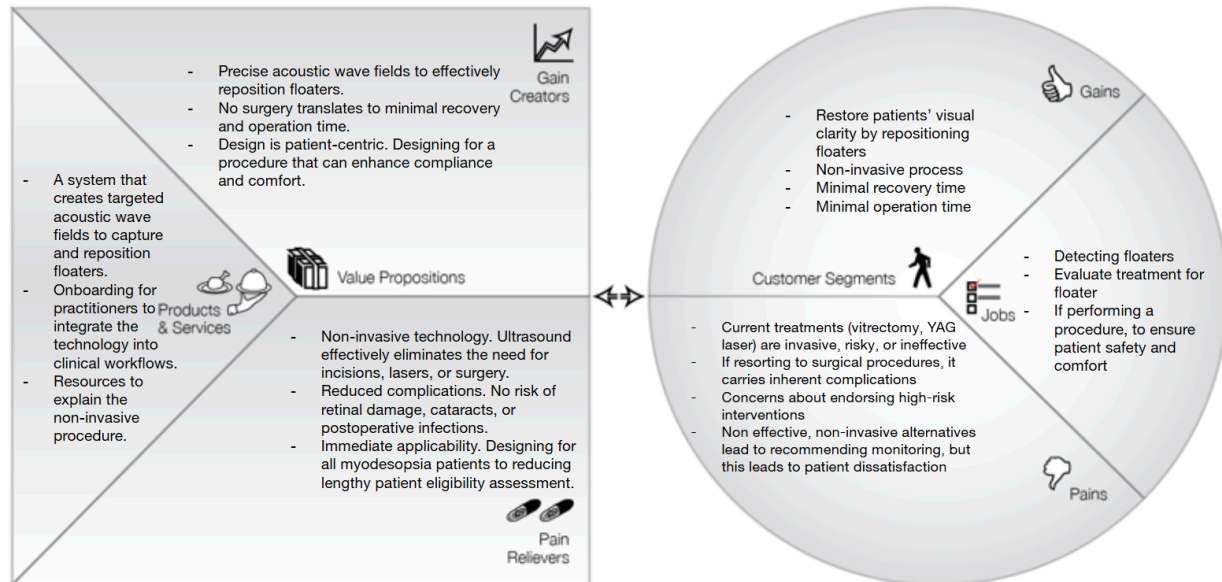


Figure 3. Value Proposition Canvas

Value Proposition Statement

With VitreaClear, you can provide effective treatment to myodesopsia patients without resorting to invasive surgery. Our ultrasound-based technology offers a precise, non-invasive, and highly controlled method for repositioning floaters, restoring visual clarity safely and efficiently.

Lean Business Model Canvas

Eye floaters significantly impair vision and quality of life for up to 60% of individuals over the age of 60. Those with myopia or Posterior Vitreous Detachment (PVD) are at particularly high risk. Current treatment options like vitrectomy, carry substantial risks, including retinal detachment and cataracts. The lack of effective, minimally-invasive alternatives leaves practitioners—optometrists and ophthalmologists—with limited treatment options and leaves patients to choose between taking these risks or living with persistent visual disturbances.

Our solution offers a non-invasive sight-saving solution by applying acoustics in a novel way. Our core ultrasound technology uses precisely arranged piezoelectric disks to generate controlled acoustic wave fields. These fields create pressure nodes that capture and reposition floaters to a non-disruptive area, restoring visual clarity without requiring incisions.

Unique Value Proposition: the treatment uses ultrasound technology to provide a safe, effective, and non-invasive alternative to existing surgical interventions.

Unfair Advantage: the support of the University of Washington in developing and patenting the technology provides credibility and establishes a strong foundation for market entry.

Key Metrics: the success of our technology will be measured through clinical and operational metrics. Critical metrics for assessing success include:

- Patient's visual clarity (NEI VFQ-25 scores)
- Rate of significant improvement
- Adoption rates among healthcare practitioners
- Patient satisfaction, measured via survey

Customer Segments: primary customer segments include optometrists and ophthalmologists, who are responsible for diagnosing and managing eye floaters. Secondary stakeholders include hospitals and eye clinics, who will assess the financial and therapeutic value of the technology. Decision-makers include clinic administrators and hospital procurement teams, who evaluate cost-effectiveness and operational feasibility.

Channels: the product will be distributed to ophthalmology clinics, optometry offices, and hospitals specializing in eye care. Collaborations with healthcare networks, professional organizations, and insurance providers will support integration and adoption.

Cost Structure: development costs include R&D, regulatory approvals, and production.

Revenue Streams: revenue will be generated through direct sales of equipment to clinics and hospitals, licensing of proprietary technology, and recurring maintenance agreements.

3.2 Market Opportunity

The ophthalmology landscape presents a compelling market opportunity driven by two critical convergent factors: the growing prevalence of eye disorders and increasing healthcare awareness. Early diagnosis and an increasing tendency for patients to seek treatment at earlier stages are driving demand for new technologies in this field. Our product is well-positioned to capitalize on the growing focus and demand for eye care technologies, particularly as patients seek more effective methods for reducing visual disturbance caused by floaters.

The ophthalmology market is experiencing rapid growth driven by increased demand for diagnostic tools and therapies, supported by rising healthcare investments and awareness. The global ophthalmology market size is valued at \$77.10 billion in 2024 and is projected to grow at a compound annual growth rate (CAGR) of 6.4%, reaching \$144.04 billion by 2034 [13].

While the market offers significant opportunities, it also presents challenges. Stringent regulatory requirements complicate market entry, requiring substantial investments in time and resources. However, these challenges can be strategically reframed as opportunities to establish market credibility, demonstrate commitment to patient safety, and differentiate through rigorous compliance.

Our primary focus will be on clinics specializing in the treatment of myodesopsia, including those serving vitrectomy candidates—a demographic with urgent needs for effective floater management. Their well-defined challenges—persistent visual disturbances and heightened surgical risk—create strong demand for non-invasive treatments with minimal or no side effects. Focusing on a solution that can offer visual clarity to this group provides an impactful entry point for introducing our innovation.

Based on the considerations outlined, we have identified three complementary market entry strategies:

1. Establishing a full-fledged business entity focused on the design, manufacturing, and direct sales of our eye floater treatment device. This approach would prioritize scaling production and creating long-term relationships with ophthalmology clinics and healthcare providers. Success would depend on securing early investments, obtaining regulatory approvals, and building a strong sales infrastructure.

2. Developing the product to commercialization and planning a strategic sale to an established ophthalmology device manufacturer or healthcare company. This approach would benefit from the broader market knowledge and distribution networks of

established firms, enabling faster market penetration and reducing operational challenges for our team.

3. Focusing on developing and securing the intellectual property rights for our product. This approach would involve licensing the IP, helping to mitigate operational and financial risks associated with manufacturing and marketing, while ensuring effective market entry for our innovation.

By strategically navigating market dynamics, regulatory landscapes, and strategic partnerships, we have the potential to effectively position our solution within the rapidly evolving ophthalmology market.

4. Approach

4.1 Evaluation of Prototype

Water-Based Prototype Testing

Methods: Initial prototypes used single-element transducers operating at 1.007 and 1.001 MHz to demonstrate acoustic particle manipulation in air. Testing was conducted at the Center for Industrial and Medical Ultrasound using a vector network analyzer (AIM 4170D Array Solutions) for impedance characterization and calibration. Open-Short-Load (OSL) calibration was performed to establish accurate measurement reference planes and correct for systematic errors. The water-based test bench addresses the acoustic impedance and wave propagation challenges in liquid media. Testing involves impedance matching and force measurement in aqueous environments. Two wiring configurations were evaluated: screw-in terminal BNC connectors with electrical tape isolation and direct BNC coaxial cable connections.

Results: Water-based prototypes successfully demonstrated acoustic particle manipulation using single-element transducers. Four different device configurations were tested, with epoxy matching layers showing optimal energy transmission characteristics and minimal reflected energy. Force measurements demonstrated particle manipulation capabilities, with measured forces ranging from 0.010g to 0.043g at power levels between 1.43W and 4.04W. Coaxial cable configurations showed improved energy transfer compared to individual wire connections. Force

measurements in water validated the acoustic manipulation concept in liquid environments more similar to vitreous conditions.

Discussion: The water-based testing validates the core acoustic manipulation concept and provides baseline performance metrics. The success of epoxy matching layers in reducing energy reflection establishes design parameters for more complex media. These results confirm that acoustic forces can effectively displace particles when properly configured.

Design Considerations

The proposed design has many potential challenges and the proposed plan is likely to deviate from the path laid out here. It is possible that technical oversights will provide additional unforeseen problems, but the main anticipated issues are the increased speed of sound through water, which results in smaller pressure nodes, along with difficulties in developing a circuit that can impart ultrasound without refraction into water as the team does not have experience with manufacturing waterproofed circuits. In order to get around these issues, it may be beneficial to utilize ultrasound gel pads or polydimethylsiloxane (PDMS) to isolate the transducers and their electrical connections from the water. This approach provides a more realistic model since it is likely that ultrasound will need to travel through a gel or pad to be transmitted into the body.

The troubleshooting necessary to achieve 3D manipulation of particles is expected to be significant as most acoustic manipulation of particles is conducted in controlled microchips. Acoustic manipulation of particles in the aqueous humor of the eye has been achieved [15], however, acoustic manipulation of particles in the vitreous has not yet been demonstrated in the literature. A combination of the novelty and technological complexity of the technology provide significant challenges for the R&D stages of this project.

4.2 Intellectual Property

Existing patents for vitreous floater treatments primarily focus on laser-based ablation techniques, like the YAG laser vitreolysis. The YAG laser vitreolysis is a non-invasive procedure using nanosecond laser pulses to vaporize floaters by converting their molecules into gas. While it avoids surgical invasiveness, it carries risks like increased symptomatic floaters and retinal damage. Its effectiveness depends on floater characteristics, and the treatment often requires multiple laser applications, making it time-consuming and inconsistent in outcomes.

Vitreaclear's ultrasound system differentiates it from existing patents and has the potential to provide a more targeted and precise alternative to YAG laser treatments, reducing risks and increasing efficacy. Our use of acoustic wave technology for floater repositioning represents a novel application in ophthalmology. Unlike prior solutions that break up or remove floaters, this method repositions them safely and effectively—a concept not evident in existing patents. This approach provides:

1. **Improved Safety:** Eliminates risks associated with laser misapplication, retinal damage, and surgical intervention.
2. **Enhanced Precision:** Targeted acoustic wave fields allow for controlled floater repositioning, minimizing collateral tissue effects.
3. **Patient Comfort:** A completely non-invasive, painless alternative with minimal recovery time.
4. **Broader Applicability:** Suitable for a wider range of myodesopsia patients.

Provisional Patent Application Filing

Recognizing the innovative nature of this approach, we have taken steps to protect our intellectual property by filing a provisional patent application titled "MANIPULATION OF EYE FLOATERS THROUGH UTILIZATION OF ACOUSTIC TWEEZERS" on June 2, 2025. The abstract of the application states: "Our proposed ultrasound-based floater repositioning system offers a non-invasive approach that uses acoustic wave technology to capture and reposition floaters out of the line of vision. By eliminating the need for surgery or laser ablation, this technique has the potential to significantly improve patient outcomes, minimize complications associated with surgical interventions, and improve overall treatment accessibility." This filing secures an early priority date for VitreaClear and provides a foundation for future non-provisional patent applications while we continue to refine and validate the technology. Additional details regarding the provisional patent application, including relevant excerpts, are provided in the appendix under "Provisional Patent Application Excerpts" for further reference.

4.3 Regulatory

Vitreaclear is a non-invasive wearable device designed to treat symptomatic vitreous floaters by relocating them out of the visual field using acoustic standing wave technology. It targets patients suffering from myodesopsia (floaters), reducing visual disturbances and improving quality of life by trapping and moving floaters away from the line of vision.

Currently, no predicate devices exist that use acoustic standing wave technology for treating eye floaters. Existing treatments, such as pars-plana vitrectomy (PPV) and YAG

laser vitreolysis, are invasive and can carry significant risks, including collateral damage. VitreaClear's approach differs significantly, as it uses ultrasound to relocate floaters without causing tissue damage. While **21 CFR 886.5350** applies to ultrasound cyclodestructive devices (used for tissue ablation in glaucoma treatments), VitreaClear's mechanism of action—relocating floaters rather than ablating tissue—precludes these devices from serving as suitable predicates. Given the lack of a direct predecessor in the FDA's database, VitreaClear will pursue the **De Novo pathway** for approval.

Class II Classification: Given its non-invasive nature and use of ultrasound technology, VitreaClear is anticipated to be classified as a Class II medical device, similar to YAG laser treatment for eye floaters.

Regulatory Pathway: Despite initial research on predicate devices, no suitable matches were found, so VitreaClear will follow the De Novo pathway for approval. This requires extensive data to demonstrate safety and effectiveness.

Development Plan:

- **Bench Testing:** The immediate focus is on bench testing to validate the acoustic standing wave mechanism. This includes experiments to generate pressure nodes, trap and relocate particles mimicking eye floaters, and test the device in a gel-like medium that simulates the eye's internal structure.
- **Future Testing:** While animal studies are not feasible at this stage, in vitro tests on porcine ocular tissue samples will inform future studies and help refine exposure limits for in vivo testing. Clinical validation is a longer-term milestone.

Sterilization and Hygiene: Since VitreaClear is a wearable device not in direct contact with the eye's internal structures, it does not require sterilization. Design considerations include cleanable surfaces and disposable components for skin-contact areas to ensure hygiene and reduce infection risk.

Safety and Compliance Considerations:

- **Electrical Safety:** Compliance with **IEC 60601** standards for medical electrical equipment is required.
- **Mechanical Safety:** The wearable design will undergo testing for comfort, stability, and fit across diverse patient populations.
- **Software Validation:** If the device incorporates software for controlling the acoustic field or monitoring performance, it will undergo validation to meet FDA guidelines for software-containing medical devices.

As we continue development, we are prepared to address ongoing regulatory, design, and testing requirements to ensure compliance and support VitreaClear's safety and efficacy.

5. Additional Considerations

In bringing a medical device to market, numerous factors can significantly impact its success. For example, the FDA regulatory pathway chosen for any treatment or device will directly influence its time-to-market and revenue generation potential. For our solution, we will likely pursue the Class II FDA approval pathway. Given the novelty of the proposed technology, regulatory and reimbursement hurdles are likely to be a significant challenge in developing and commercializing this technology.

Our goal in developing this product is to benefit society by addressing a condition that adversely affects the quality of life of millions in the United States. By restoring clear vision to patients with no or minimal side effects, we aim to significantly improve access to effective treatments for eye floaters. A core focus of our mission is to ensure the treatment is accessible and equitable to maximize its positive impact on public health.

In addition, our team is dedicated to environmental sustainability. We are committed to minimizing the carbon footprint and waste associated with our prototyping and manufacturing processes. This includes implementing waste-reduction strategies throughout the product lifecycle. To further reduce environmental impact, we plan to use porcine eyes from animals already sacrificed for other research purposes. This approach allows us to make full use of available resources while adhering to ethical and sustainable practices.

6. Conclusion and Future Work

Eye floaters are a significant unmet medical need, affecting 60% of individuals over 60 and impairing daily activities like reading and driving. Current treatments, from invasive vitrectomy with risks of retinal detachment to inconsistent YAG laser vitreolysis, fail to offer safe, accessible solutions. The economic burden of visual impairment in the U.S. reaches \$16.33 billion annually, yet floater-specific therapies remain underdeveloped. With the global ophthalmology market at \$77.10 billion in 2024, growing at a 6.4% CAGR to \$144.04 billion by 2034, there is a substantial opportunity for a non-invasive solution to improve health, enhance convenience, and reduce costs [5].

VitreaClear is an ultrasound-based acoustic repositioning system that non-invasively repositions vitreous floaters using controlled acoustic wave fields from piezoelectric

transducers. Strengths include its non-invasive design eliminating surgical risks, precise floater manipulation, minimal side effects per FDA guidelines, scalability across patient profiles, and real-time monitoring potential. Early prototypes showed particle manipulation in aqueous media, achieving forces of 0.010g to 0.043g at 1.43W to 4.04W power levels.

Focus on transitioning to phased array transducer configurations for controlled pressure fields, integrating thermocouples for real-time temperature monitoring, enabling mechanical actuation for dynamic pressure field mobilization, and incorporating ultrasound imaging for precision targeting are the technical development priorities. In this development process, we expect to dedicate significant time to tackle the technical complexity in developing 3D transducer configurations in viscous media.

As such, at a high level, the team's testing and validation roadmap includes advancing testing in viscous fluids mimicking vitreous conditions, followed by cadaver eye validation for anatomical relevance, then progressing toward clinical trials to confirm safety and efficacy in humans. These steps not only support technical development but are also important for regulatory data collection. Furthermore, the team is seeking further engagement with clinicians and regulatory consultants. Building on progress, the VitreaClear team is committed to our vision of delivering a safe, effective, and accessible solution, transforming floater management for myodesopsia patients and enhancing quality of life for millions.

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Appendices

Appendix A

Table 1. Fall Quarter: Initial and Refined Need Statements

Initial	Refined
A way to alleviate visual discomfort from eye floaters in myopic patients, minimizing the need for invasive procedures and improving overall visual comfort.	A way to alleviate visual discomfort caused by vitreous floaters in patients with myopia so that they can improve their visual quality and reduce the need for invasive interventions.
A non-invasive approach to managing recurring floaters in post-vitrectomy patients to improve patient adaptation.	A non-invasive solution for managing recurrent floaters in post-vitrectomy patients so that they can adapt better to the condition and avoid further complications.
An approach to lessen disturbances from eye floaters in aging patients whose symptoms impair daily activities, ensuring treatments match symptom severity.	A method to improve visual clarity for aging patients with posterior vitreous detachment (PVD) so that they experience fewer disturbances without requiring invasive treatment.

Appendix B

Table 2. CTC and CTQ Scores

Critical to Customer (CTC)	Patient (0.75)	Optometrist (1)	Corresponding CTQ
Comfort	9	1	Comfort Scale (1-10)
Non-invasive	9	3	Does it pierce the body?
No Side Effects	9	9	# of side effects
Feedback & Monitoring	1	9	Refresh rate (during procedure)
Effectiveness of solution	9	9	The National Eye Institute's Visual Function Questionnaire (NEI VFQ-25)
Little setup and prep work	1	9	How long it takes to set up the procedure (in minutes)

Appendix C

Full Matrix (enlarged sections provided on the following page)

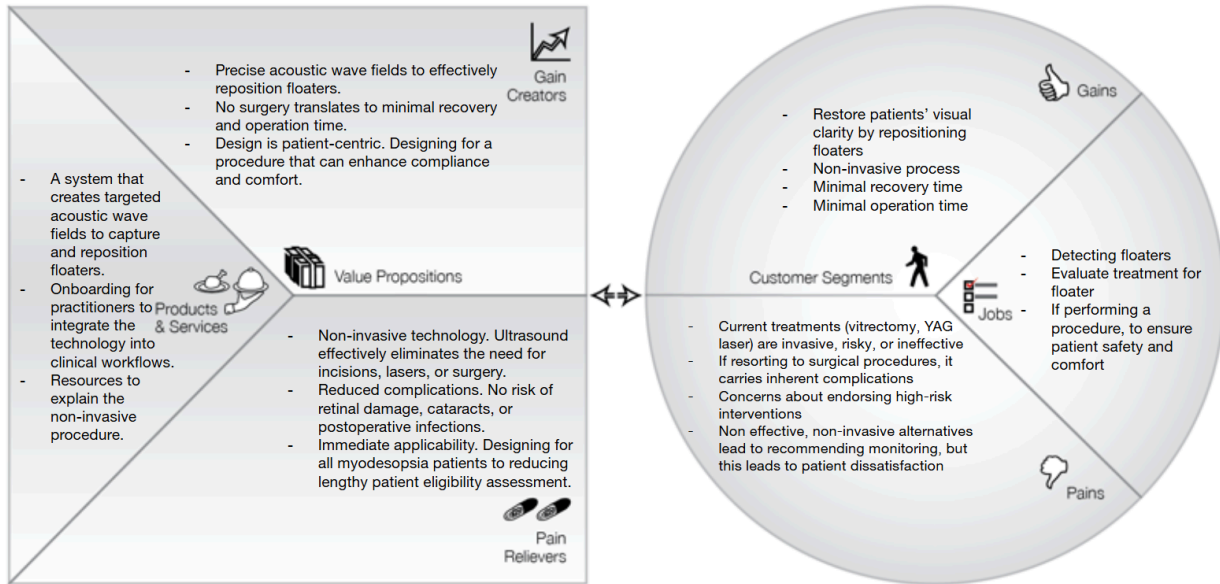
Vitreaclear				Design Specifications (CTQs)								COMPETITIVE BENCHMARK								
A way to safely and precisely reposition visually disruptive vitreous floaters for myopia patients so that they can achieve improved vision without undergoing invasive surgery or risking retinal complications.				#1	#2	#3	#4	#5	#6	#7		Monitoring and Passive Adaptation	YAG Laser Treatment	Vitreotomy	Automated YAG Laser Treatment	VR Therapy	Superior Vitreous Fluid	Floater Dissolving Eye Drops	Vibration	Acoustic Repositioning
Relative weight of customer	1.00	1.00	PRIORITY SCORE	Comfort Scale (1-10)	Does it pierce the body?	# of side effects	Refresh rate (during procedure)	NEI VFQ-25	Duration of procedure and set-up	# of follow-up appointments										
Critical to Customer (CTC)	Patient	Clinician																		
Patient Comfort	9	3	12	9	9	3	1	1	1	3		9	3	1	9	9	1	3	3	9
Non-invasive	9	3	12	3	9	9	1	1	1	3		9	9	1	9	9	1	9	9	9
Minimal Side Effects	9	9	18	3	3	9	1	3	1	3		9	3	1	9	3	1	3	3	9
Live Feedback During Operation	3	9	10	1	1	1	9	1	3	3		1	3	9	9	3	9	3	9	9
Patient Satisfaction	9	9	18	3	3	3	1	9	3	3		1	3	3	3	9	9	3	3	9
Little Set-up and Prep Work	1	9	10	1	9	1	3	1	9	2		9	3	1	9	9	1	9	3	2
Efficient Patient Follow-up	3	9	12	1	1	1	1	3	1	9		3	3	1	3	3	1	3	3	3
CTQ IMPORTANCE				284	436	392	192	296	228	224										
TECHNICAL BENCHMARKING	11404	No Intervention		9	9	9	1	1	1	3	2									
	8772	Laser		3	9	3		3	3	3	2									
	4180	Vitreotomy		1	1	1		9	3	1	2									
	13404	Automated Yag		9	9	3		9	9	3	3									
	12996	VR Therapy		9	9	3		9	3	9	9									
	5956	Vitreous Fluid		1	1	1		9	9	1	3									
	11916	Eye Drops		3	9	3		3	9	9	3									
	9924	Vibration		3	9	3		9	3	3	3									
	15756	Acoustic Repositioning		9	9	9		9	9	3	3									
				532	348	208	648	480	316	516	408	696								

VitreaClear				Design Specifications (CTQs)																	
A way to safely and precisely reposition visually disruptive vitreous floaters for myodesopsia patients so that they can achieve improved vision without undergoing invasive surgery or risking retinal complications.				#1	#2	#3	#4	#5	#6	#7											
				Critical to Customer (CTC)	Patient	Clinician	PRIORITY SCORE	Comfort Scale (1-10)	Does it pierce the body?	# of side effects	Refresh rate (during procedure)	NEI VFQ-25	Duration of procedure and set-up	# of follow-up appointments							
Relative weight of customer	1.00	1.00																			
Patient Comfort	9	3	12												9	9	3	1	1	1	1
Non-invasive	9	3	12												3	9	9	1	1	1	1
Minimal Side Effects	9	9	18												3	3	9	1	3	1	1
Live Feedback During Operation	1	9	10												1	1	1	9	1	3	1
Patient Satisfaction	9	9	18												3	3	3	1	9	3	3
Little Set-up and Prep Work	1	9	10												1	9	1	3	1	9	1
Efficient Patient Follow-up	3	9	12												1	1	1	1	3	1	9
CTQ IMPORTANCE				284	436	392	192	296	228	224											
TECHNICAL BENCHMARKING	11404	No Intervention	9	9	9	1	1	3	1												
	8772	Laser	3	9	3	3	3	3	3												
	4180	Vitrectomy	1	1	1	9	3	1	1												
	13404	Automated Yag	9	9	3	9	9	3	3												
	12996	VR Therapy	9	9	3	9	3	9	3												
	5956	Vitreous Fluid	1	1	1	9	9	1	1												
	11916	Eye Drops	3	9	3	3	9	9	3												
	9924	Vibration	3	9	3	9	3	3	3												
15756	Acoustic Repositioning	9	9	9	9	9	3														

COMPETITIVE BENCHMARK								
Monitoring and Passive Adaptation	YAG Laser Treatment	Vitrectomy	Automated YAG Laser Treatment	VR Therapy	Superior Vitreous Fluid	Floater Dissolving Eye drops	Vibration	Acoustic Repositioning
9	3	1	9	9	1	3	3	9
9	9	1	9	9	1	9	9	9
9	3	1	9	3	1	3	3	9
1	3	9	9	3	9	3	9	9
1	3	3	3	3	9	9	3	9
9	3	1	9	9	1	9	3	3
3	3	1	3	3	1	3	3	3
532	348	208	648	480	316	516	408	696

Appendix D

Value Proposition Canvas



Appendix E

Lean Canvas		VitreaClear		3/15/24
				Iteration #2
Problem 1. High prevalence (Eye floaters impact up to 60% of individuals over the age of 60.) 2. Limited treatment options (While treatment options like vitrectomy are available, with approximately 225,000 vitrectomies performed each year, this procedure carries significant risks like retinal detachment and cataracts.) 3. Lack of non-invasive solutions (Optometrists primarily recommend monitoring as the main approach to managing eye floaters, as no non-invasive solutions are currently available. This forces patients to endure visual disturbances without any practical alternatives.)	Solution 1. <u>Non-invasive</u>: reduces retinal complications associated with invasive surgery 2. Use of <u>ultrasound</u>-based technology to target, capture, and reposition floaters. 3. <u>Acoustic manipulation</u> of particles has the potential to control and arrange floaters without causing damage to ocular tissue.	Unique Value Proposition Compared to existing solutions for treating eye floaters which rely on invasive surgeries and therefore carry significant risks of retinal complications, our technology provides a non-invasive approach that maintains effectiveness and ensures a safer treatment outcome.	Unfair Advantage • Proprietary technology (UW-supported patents)	Customer Segments • Primary customer: optometrists • Purchasers: hospitals, eye clinics, vision care centers • Decision-makers: administrators, medical directors, ophthalmology department heads
	Key Metrics Clinical Success: • Visual clarity measured by NEI VFQ-25 scores • Percentage of floater successfully treated Adoption Rate: • Number of clinics adopting the technology		Channels • Partnerships with ophthalmology clinics, optometry offices, and vision centers • Direct engagement with hospitals and insurance networks	
Cost Structure • Production and design costs • IP and legal costs • Regulatory compliance (costs for FDA approval and certifications) • Marketing		Revenue Streams • Licensing fees • Device sales to clinics, hospitals • Maintenance/warranty contracts		
PRODUCT		MARKET		

Appendix F

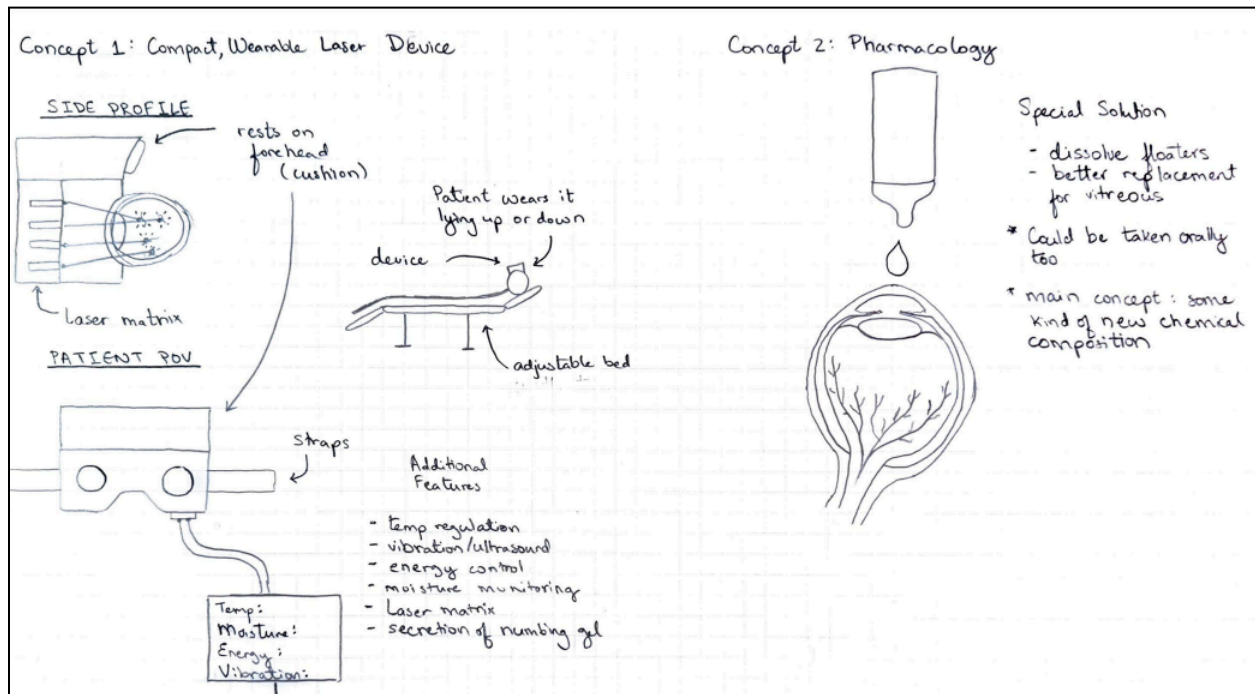


Figure 2. Drawings of combined concepts

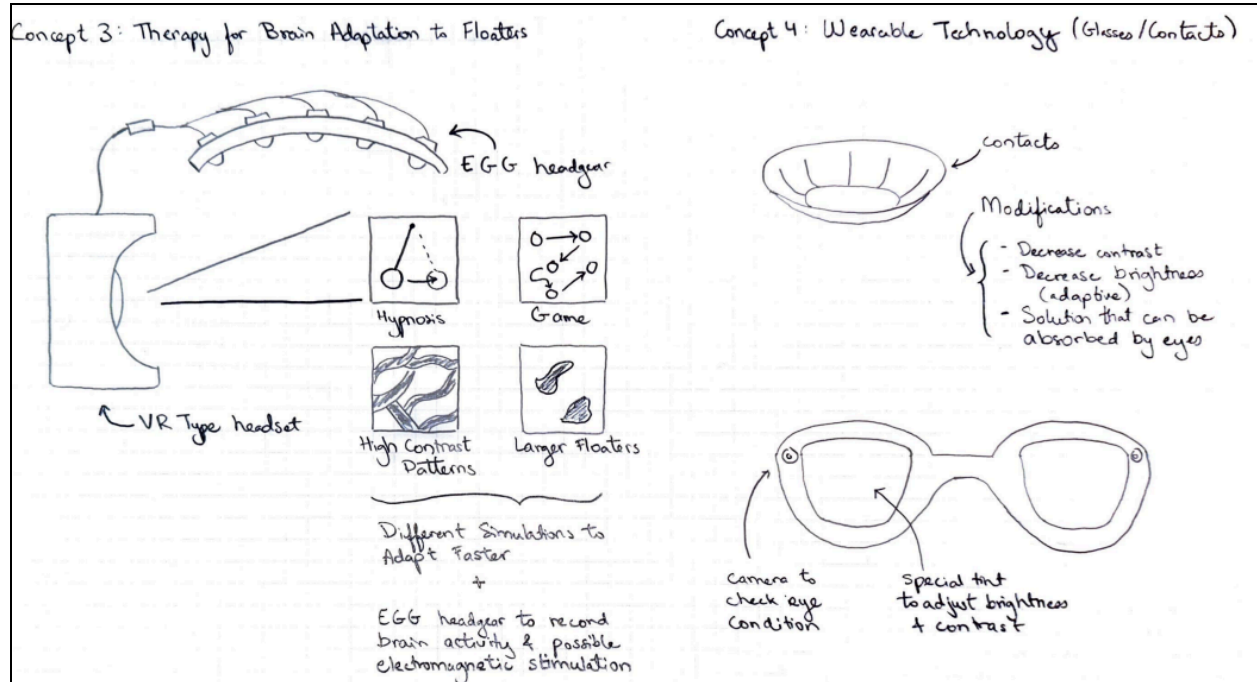


Figure 3. Drawings of combined concepts

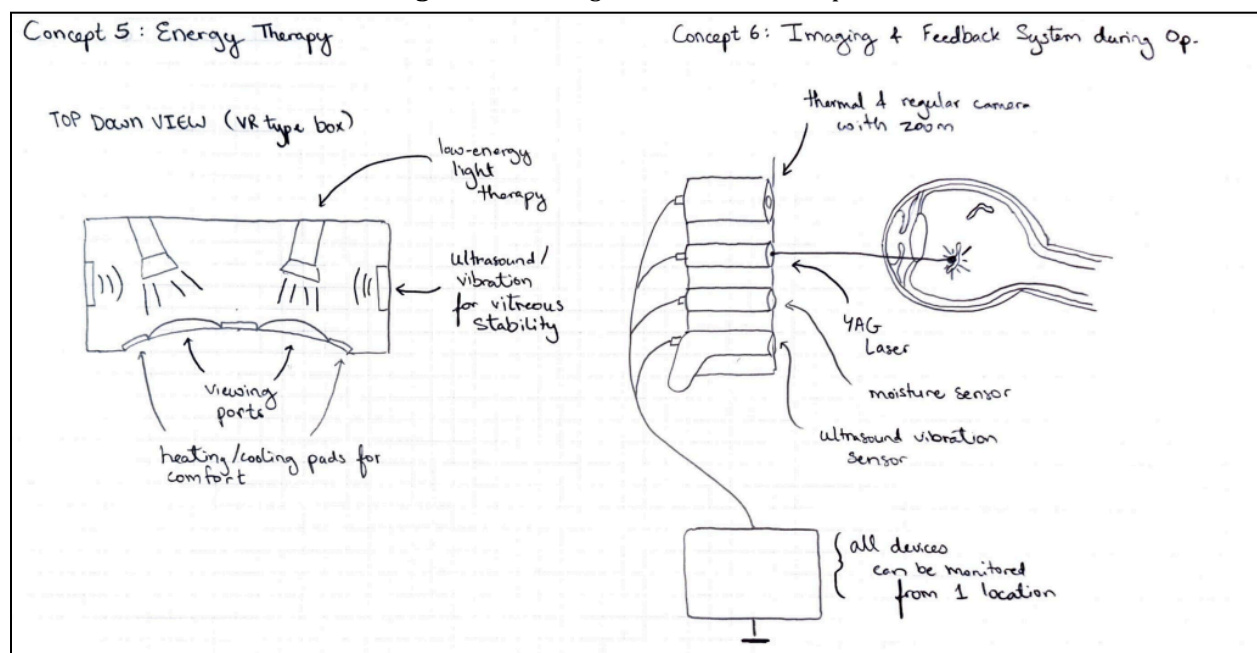


Figure 4. Drawings of combined concepts

Appendix G

High Intensity Focused Ultrasound (HIFU) Beam

The "HIFU Beam" simulator, recommended by Dr. Thomas Matula, Director of the Center for Industrial and Medical Ultrasound (CIMU) at the University of Washington's Applied Physics Lab, is a tool for predicting axially symmetric nonlinear acoustic fields generated by focused transducers in layered media. It was investigated as part of our research into acoustic wave technologies. As a preface, while this simulator provided theoretical insights into nonlinear acoustic propagation, it ultimately did not inform the actual design of VitreaClear due to differences in application focus. Below is a condensed summary of relevant concepts, parameters, and models from the simulator that were explored during the project.

Concepts in Nonlinear Acoustic Propagation

Nonlinear Acoustic Propagation: Occurs at high amplitudes where sound waves distort, generating harmonics and potentially forming shock waves.

Acoustic Radiation Forces (ARF): A steady force resulting from momentum transfer in acoustic waves, proportional to the energy density gradient. ARF is essential for trapping and moving particles (such as floaters) in focused or standing wave fields, especially in the focal region where spatial and temporal pressure structures enhance manipulation efficiency.

Shock Wave Effects: Formation of shock fronts with steep pressure gradients enhances local forces and acoustic streaming. Improves particle capture and movement capabilities.

Simulation Models

The simulator uses two primary nonlinear acoustic propagation models relevant to our transducer design:

1. **Khokhlov-Zabolotskaya-Kuznetsov (KZK) Equation:** Models radially symmetric beams using a parabolic approximation for diffraction, accounting for nonlinearity, diffraction, and absorption in a thermo-viscous medium. Suitable for simulating focused ultrasound beams with moderate focusing.
2. **One-Way Westervelt Equation (WAPE):** Models radially symmetric beams with a wide-angle Padé approximation for diffraction, offering improved accuracy for strongly focused beams by solving only forward-propagating waves and ignoring reflections.

Simulation Parameters

- Medium Properties (Water as Baseline):
- Sound Speed (c_0): 1500 m/s
- Density (ρ_0): 1000 kg/m³
- Nonlinearity Coefficient (β): 3.5
- Diffusivity (δ): 4.33 mm²/s

Absorption Model: Frequency-dependent absorption with thermoviscous effects in water (minimal) and power-law absorption in tissue-like media (relevant for future ocular simulations).

Transducer Design Parameters (Example for Single-Element Spherical Bowl):

- Diameter: 100 mm
- Focal Distance: 90 mm
- Operational Frequency: 1 MHz

Annular Array Consideration: Arrays with multiple elements, which allow electronic steering of focus via phase delays. This may be relevant for future phased array development in VitreaClear.

Source Pressure Amplitudes and Nonlinear Regimes:

- Linear: 0.015 MPa (minimal distortion)
- Quasi-linear: 0.15 MPa (harmonics emerge)
- Developed Shock: 0.45 MPa (sharp pressure changes)
- Saturated Shock: 0.7–1.45 MPa (absorption limits growth)

Simulation Domain and Numerical Parameters:

- Grid Steps: Radial (0.0025–0.0125 mm), Axial (0.0125–0.08 mm) for high resolution
- Maximum Harmonics (N_{max}): 1000 (capturing nonlinear effects)

Perfectly Matched Layer (PML): absorbs outgoing waves to prevent boundary reflections

While acoustic radiation force and focused ultrasound fields are highly relevant to particle manipulation in this project, the HIFU beam simulator's focus on high-intensity focused ultrasound diverged from VitreaClear's low-intensity floater repositioning goals. Consequently, it did not directly inform our design but enriched our understanding of acoustic principles guiding our approach.

Appendix H

Test Plan

Introduction

The following test plan outlines procedures to validate the proof-of-concept for an acoustic manipulation system designed to push or engage with a neutrally buoyant particle in a water medium. The prototype uses a 5.5-gallon fish tank, a matching layer between a 1–3 MHz transducer and a 3D-printed lens, and a rubber absorber at the tank bottom. The primary objectives are to:

1. Demonstrate the system's ability to directionally move a neutrally buoyant particle using standing ultrasound waves.
2. Establish a structured test bench for future quantitative studies.

Measurements will include particle displacement, transducer output, water temperature, and reflected acoustic power.

Test Constraints

Constraint Name	Description	Value/Range
Tank Volume	Volume of water tank	5.5 gallons (fixed)
Medium	Water	Degassed, room temp
Acoustic Frequency	Transducer operational range	1–3 MHz (per transducer spec)
Reflected Power	Acceptable reflected power at interface	$\leq 6\%$
Matching Layer	Placement/material between element & lens	As per design, no air bubbles

Test Parameters

Parameter Name	Description	Range/Notes
Particle Size	Neutrally buoyant particle* diameter	425-500 μm

Transducer Voltage	Input to transducer	As per amplifier spec, record value
Lens Position	Distance from transducer to water surface	Fixed per setup, record value
Water Temperature	Measured before each test	20–25°C (room temp)

*Particles used are fluorescent polyethylene microspheres

Test Variables

Variable Name	Description	Values/Range
Frequency (f)	Acoustic frequency applied	[1][2][3] MHz
Input Power (P)	Electrical power to transducer	1–10 W
Particle Material	Type of neutrally buoyant particle	-
Particle Position	Initial position in tank	-

Measurements

Measurement Name	Description	Units/Range
Particle Displacement	Distance moved by particle during test	mm/cm
Input Power	Power delivered to transducer	W
Output Frequency	Frequency measured at transducer	MHz
Water Temperature	Temperature of water during test	°C

Reflected Power	Power reflected at matching layer	% ($\leq 6\%$ target
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Appendix I

Test Procedures

Test Objective

Verify through controlled testing that the acoustic manipulation prototype—consisting of a 1–3 MHz transducer, matching layer, 3D-printed lens, and rubber absorber in a 5.5-gallon water tank—can reliably generate sufficient acoustic force to move a neutrally buoyant particle in water, while ensuring reflected power at the matching layer does not exceed 6% as per CIMU lab standards. Collect quantitative data on particle displacement, reflected power, and system operating parameters to establish proof of concept and create a structured test bench for future experiments.

Equipment Required

Qty	Description	Specs/Calibration
1	Fish tank	5.5 gallons, glass/acrylic, clean
-	Transducers	1–3 MHz Part #
5	3D-printed lens	As per CAD
1	Rubber absorber	Covers tank bottom
-	Neutrally buoyant particles*	425-500 μm , density matched to water
-	Sensors (measure power reflected at matching layer)	% ($\leq 6\%$ target)

*Particles used are fluorescent polyethylene microspheres

Test Procedure

1. Setup & Safety Checks

1.1 Place fish tank on stable, vibration-isolated table.

OK? _____

1.2 Fill tank with degassed water to marked fill line (5.5 gal).

OK? _____

1.3 Install rubber absorber at tank bottom, ensuring full coverage.

OK? _____

1.4 Mount transducer and lens assembly above tank, align with tank center.

Alignment error ≤ 2 mm: _____ mm

1.5 Attach matching layer between transducer and lens, check for air bubbles.

No visible bubbles? OK? _____

1.6 Connect function generator, amplifier, oscilloscope, and power meter as per circuit diagram.

OK? _____

1.7 Place temperature sensor in water, record initial temperature.

Measured: _____ °C

1.8 Insert neutrally buoyant particle at designated start position.

OK? _____

1.9 Confirm all electrical connections are secure and dry.

OK? _____

2. Power Up

2.1 Set function generator to target frequency (e.g., 1 MHz).

Set: _____ MHz

2.2 Set amplifier to desired input power.

Set: _____ W

2.3 Turn on power supply, verify system voltage within safe range.

Voltage: _____ V

2.4 Power meter reads baseline (no signal) reflected power.

Measured: _____ W

3. System-Specific Tests

3.1 Activate ultrasound, observe particle movement.

OK? _____

3.2 Measure and record particle displacement after 10 s.

Displacement: _____ mm

3.3 Measure reflected power at matching layer.

Reflected Power: _____ %

3.4 Repeat for each frequency (1, 2, 3 MHz) and input power (1, 5, 10 W).

OK? _____

3.5 Record water temperature at end of each run.

Measured: _____ °C

4. Shut Down

4.1 Turn off amplifier and function generator.

OK? _____

4.2 Disconnect power supply.

OK? _____

4.3 Remove and store transducer and lens assembly.

OK? _____

4.4 Remove particle(s), clean tank and components.

OK? _____

4.5 Store all equipment as per CIMU lab protocol.

OK? _____

Appendix J

Provisional Patent Application

A provisional patent application titled "MANIPULATION OF EYE FLOATERS THROUGH UTILIZATION OF ACOUSTIC TWEEZERS" was filed on June 2, 2025, under reference number 3915-P1409US.PRO.UW. To maintain confidentiality of the application, a representative claim is included below to illustrate the scope of the technology:

Representative Claim:

"A method of repositioning vitreous floaters in an eye of the patient, the method comprising: positioning an ultrasound device proximate to the eye, wherein the ultrasound device comprising piezoelectric ultrasound transducers; transmitting ultrasound waves to the eye; forming ultrasound force field at least partially configured around the floaters in the eye; and repositioning the floaters by manipulating the ultrasound force field."

Appendix K CAD Design

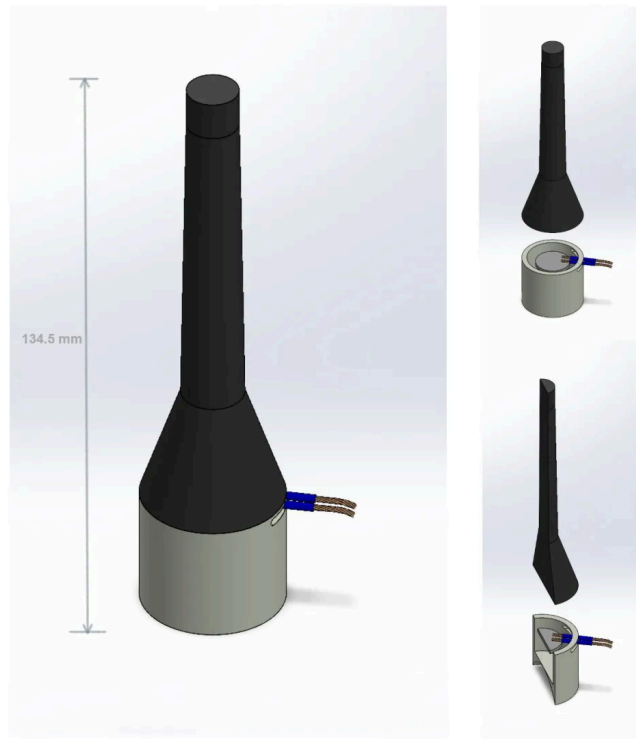


Figure K. Transducer and Device Assembly

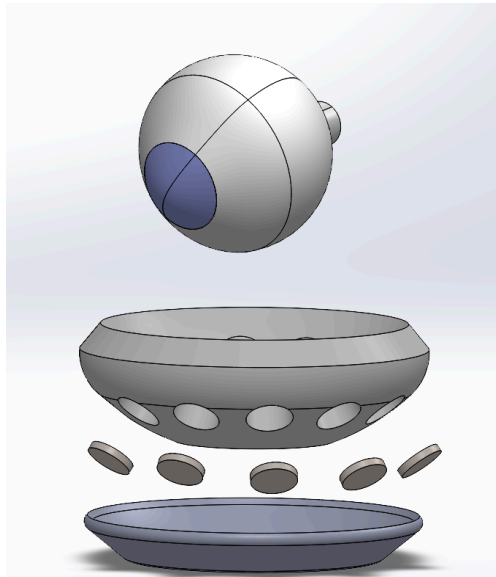


Figure L. Future Transducer Array and Device Assembly